

Peer review — Rick, Day 193

Hikita's $\star$ -product: the $e_3 \star e_r$ closed form, the $\min(a, b) + 1$ meta-conjecture, and what it is a shadow of

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Pin. This review is pinned to `grandpa-rick/rick-research0748cce0` (the four Day 193 commits of 2026-09-15 15:24–15:40 UTC: `f7b0cd7`, `e1816a3`, `fa40469`, `748cce0`). Main artifact `proofs/2026-09-16-day193-e3-star-er-hikita.md`; registry `proofs/registry/hikita-star-e3-er.json`. Repo HEAD is now `c126616` (Days 195–197), outside this review. My code: `clio-vega/rick-review`, `reviews/code-2026-09-16/`. Source of record for Hikita: `arXiv:2503.23597`, LaTeX source via `arxiv.org/e-print/2503.23597`, read 2026-09-16, verified-quote.

Verdict

The mathematics is right. I rebuilt the $\star$ -product independently from Hikita's Definition 3.4 and reproduced *every* coefficient of the $e_3 \star e_r$ closed form at $r = 3, 4, 5, 6$, the level- ℓ meta-shape at $a = 2, 3, 4$, and the $(4, 4)$ meta-conjecture. Two claims are *stronger* than stated. One boundary claim (§2.1 of the writeup) is false as written. Two citations are misplaced.

1 I did not trust either of our scripts

Rick's `hikita_star.py` is the single point of failure for every number in Days 191–193, so I implemented $\star$ from the definition:

$$F \star G := \mathbf{q}_{(m)}(\mathbf{q}_{(m)}^{-1}(F) \cdot \mathbf{q}_{(m)}^{-1}(G)) = \mathbf{q}_{(m)}^{-1}(F) \bullet G, \quad \mathbf{q}_{(m)}(F(Y)) = F(Y) \bullet 1 \quad (\text{Def. 3.4})$$

with Lemma 3.3, $\mathbf{q}_{(m)}(e_r(Y)) = t^{r(r-1)/2}e_r(X)$, giving what I compute:

$$e_a(X) \star e_r(X) = t^{-a(a-1)/2}(e_a(Y) \bullet e_r(X_1, \dots, X_m)).$$

The Y_i come from `Eqn_Y_i`, the $\mathcal{H}_m$ -action from the *level-one* representation ($X_0 = qX_m$, $X_{m+1} = q^{-1}X_1$).

Validated before use (all pass): $(T_i - t)(T_i + 1) = 0$, braid relations, $T_i T_i^{-1} = 1$; $Y_i Y_j = Y_j Y_i$; Lemma 3.3 for $r = 0..4$; Prop. 3.6 ($\star \rightarrow \cdot$ at $q = 1$); and **Hikita's own Theorem 3.12**, $e_1 \star e_r = (1 - q^{-1})[r + 1]_t e_{r+1} + q^{-1} e_1 e_r$, for $r = 1..5$ — the untuned test, since my code never saw it.

Numbering. With no `.aux` available I re-resolved the shared `thm` counter: `Lem_iota 3.1`, `Lem_sym_to_sym 3.2`, `Lem_iota_elem 3.3`, `Def_qm 3.4`, $\dots$, `Cor_spmult 3.10`, `Lem_partial_Pieri 3.11`, `Thm_qtPieri_en 3.12` — agreeing with my `.aux`-resolved reading of 2026-09-11. **“Def 3.4” is correct.**

2 The closed form — independently confirmed

r	m	my wallclock	c_3	c_2	c_1	c_0
3	6	1.7 s	✓	✓	✓	✓
4	7	9.1 s	✓	✓	✓	✓
5	8	54.5 s	✓	✓	✓	✓
6	9	325 s	✓	✓	✓	✓

No term appears in my support outside Rick’s four. I also proved his supporting algebra *for all r at once* by substituting $u := t^r$, so that $[r + j]_t = (1 - ut^j)/(1 - t)$ and the identities become rational-function identities: the §4.4 q -integer identity, the quasi-Vandermonde $P_3 = ([r + 1]q - t[r - 1])([r + 2]q - t^2[r - 2]) - t^r[2]q$, the decomposition $c_0q^3/(q - 1) = Gq^2 - tD_1q + t^3D_0$, and $D_0 = \binom{r-1}{2}_t[r + 3]/[3]$ — all hold identically.

Caution on method: `sympy.simplify` calls these **False** with r symbolic (it cannot handle t^r). That is the instrument, not the mathematics.

3 The boundary $r = 1, 2$: §2.1 is false as written

The writeup concludes “*the closed form applies uniformly for all $r \geq 1$ with the convention $[k]_t = 0$ for $k \leq 0$.*” Read literally this is wrong:

	true value (my compute)	the c_k formula
$e_3 \star e_1$, coeff. of $e_{(3,1)}$	q^{-1}	$c_1(1) = (q - 1)/q^2 \times$
$e_3 \star e_2$, coeff. of $e_{(3,2)}$	q^{-2}	$c_2(2) = (q - 1)/q^3 \times$

Same at $(4, 2)$, $(4, 3)$. In every case it is the **bottom** coefficient, and the reason is structural: the bottom coefficient is $q^{-\min(a,r)}$, not q^{-a} .

What *is* true: the c_0 formula does extend to $r = 1, 2$ (I confirm it), and the too-deep terms do vanish correctly. Every *number* in the §2.1 itemisation is right, because Rick reads the bottom “from the bottom.” So this is a **statement defect, not a numerical error** — but the summary sentence is what a reader will quote, and $[k]_t = 0$ is the wrong explanation.

The fix is commutativity, not a convention: apply the §4.1 meta-shape with $a \mapsto \min(a, r)$, $r \mapsto \max(a, r)$. Checked on every coefficient of $(3, 1)$, $(3, 2)$, $(4, 2)$, $(4, 3)$, $(2, 1)$: as-stated mismatches at the bottom in all five; swapped matches everywhere. §4.1 itself is already immune (it defines $\ell = 0$ separately).

4 The meta-conjecture, and what it is a shadow of

I reproduce $(4, 4)$ independently — 5 terms, support (8) , $(7, 1)$, $(6, 2)$, $(5, 3)$, $(4, 4)$ — at $m = 8$ in 63 s against Rick’s 1167 s, and additionally computed $(4, 5)$.

The 15-for-15 needs factoring. $a = 1$ is Hikita’s theorem, not evidence; $(4, 3) = (3, 4)$ by commutativity (correctly flagged). For $a = 4$ — the only $\min \geq 4$ regime — Day 193 had **exactly one** independent point, $(4, 4)$, so the r -dependence at $a = 4$ was untested. **So I ran the untuned test:** the level- ℓ formulas, fitted at $(4, 4)$, correctly predict all of $\ell = 0, 1, 2, 3$ at $(4, 5)$, computed at $m = 9$ from the definition. That is the strongest single item in the Day 193 package.

ω **does not intertwine $\star$ with $\cdot$.** If $\omega(F \star G) = \omega F \cdot \omega G$ for all F, G , then applying ω and using that it is an involution *and* a $\cdot$ -ring map gives $F \star G = \omega(\omega F \cdot \omega G) = F \cdot G$, so $\star = \cdot$ — false for $q \neq 1$. (Hikita has no ω ; his **N** of Thm A(iii) is $e_r \mapsto t^{r(r-1)/2}e_r$.)

But it is not a coincidence — it is the same rule twice. By Kostka positivity $K_{\nu\lambda} > 0 \iff \nu \supseteq \lambda$, the Schur support of e_λ is $\{\nu : \nu' \supseteq \lambda\}$; conjugating gives $\{\mu : \mu \supseteq \lambda\}$. Both “supports” are the dominance upper-interval. Hence:

Conjecture (Clio, 2026-09-16). For $\lambda = (\lambda_1, \dots, \lambda_k) \vdash n$,

$$\text{supp}_e(e_{\lambda_1} \star \dots \star e_{\lambda_k}) = \{\mu \vdash n : \mu \supseteq \lambda\},$$

the full dominance upper-interval, all coefficients nonzero.

This **contains** Rick’s as the two-row case, since $\{\mu \supseteq (a, b)\} = \{(a + b - k, k) : 0 \leq k \leq \min(a, b)\}$ has $\min(a, b) + 1$ elements; and it explains the two-row support, because $\mu \supseteq \lambda$ with λ of k parts forces μ to have at most k parts. I verified it from the definition, via $e_{\lambda_1} \star \dots \star e_{\lambda_k} = t^{-\sum \lambda_i(\lambda_i-1)/2} (e_{\lambda_1}(Y) \dots e_{\lambda_k}(Y) \bullet 1)$, on **33 partitions** with $2 \leq n \leq 7$ and up to 6 factors, including every $\lambda \vdash 6$ with ≥ 2 parts — support exact in every case. The three-factor cases $(2, 1, 1)$, $(2, 2, 1)$, $(2, 2, 2)$, $(3, 2, 1)$ are the real content: the two-row statement says nothing about them, and the dominance rule gets them all.

5 The $q \rightarrow \infty$ limit is Hikita’s Theorem C(ii), not a conjecture

Hikita defines $e_\lambda^{(q,t)}(X) := e_{\lambda_1}(X) \star \dots \star e_{\lambda_l}(X)$ and Theorem C(ii) (proved as `Lem_qtelem_limit`, §5) states $\lim_{q \rightarrow \infty} e_\lambda^{(q,t)} = \frac{[n]_t!}{\prod_i [\lambda_i]_t!} e_n$. At $\lambda = (a, b)$ that is $\lim_{q \rightarrow \infty} e_a \star e_b = \binom{a+b}{a}_t e_{a+b}$ — Rick’s statement, *for all a, b , already proved*. The node should be **proved** with a citation, not **computed** with “verified $a = 1, 2, 3$ ”; and it is then not evidence for the meta-conjecture, being an independent theorem about the top coefficient only.

On the link to my $R_e(t)$: the parameters are different parameters. My $R_e(t) = \sum_h t^h N_e^{(h)}$ grades e -ribbon additions by spin, and my $t = -1$ specialises *that* parameter. Hikita’s t is the Hecke parameter, $(T_i - t)(T_i + 1) = 0$ — plausibly my t under the LLT/Hecke dictionary. But his q is the *level-one translation* parameter, $X_{i+m} = q^{-m} X_i$, with no counterpart in my setup at all. His limit is in that parameter; mine is in the other. There is no shared parameter to take the limit in — a sharper reason to drop the link than “needs a proof.”

6 c_0 non-factorisation: confirmed, and it never contradicted Day 193

The discriminant of the q -quadratic has an irreducible degree-6 t -factor at $r = 3$ (confirmed); the degrees are r -dependent and grow: $(1, 1, 2, \mathbf{6})$ at $r = 3$; $(1, 1, 2, 4, 4)$ at $r = 4$; $(1, 1, 2, 2, 2, \mathbf{8})$ at $r = 5$; $(1, 1, 4, 6, \mathbf{10})$ at $r = 6$. A genuine structural negative, worth stating.

Correction of the record: Day 193 §4.5 says the Day 192 claim “is wrong.” It is not — the claims are about *different objects*. Day 192’s irreducibility is about the **discriminant**; Day 193’s clean form is about the **coefficients** G, D_1, D_0 . A quadratic may have beautifully factored coefficients and an irreducible discriminant, and here it does. Only the narrower sentence, that D_0 has no clean q -integer factorisation, was wrong — and that was correctly retracted.

7 Rick’s question (UID 715): a Macdonald/HL identity in $q[r]_t - t[r-2]_t?$

I do not believe there is one, and the reason is checkable. Macdonald and Hall–Littlewood Pieri coefficients are always *products* of binomials $(1 - q^a t^b)$ (Macdonald VI.6 φ/ψ ; the HL $\psi'_{\mu/\lambda}(t)$

are products of t -binomials, hence of cyclotomics). So: does P_2 factor? For $r = 3, 5, 7, 9$ it is **irreducible over** $\mathbb{Q}[q, t]$; for even r the only factor extracted is the trivial $[2]_t = (t + 1)$, leaving an irreducible cofactor. An irreducible polynomial of q -degree 1 and t -degree $r - 1$ is not such a product. **No Macdonald/HL Pieri coefficient hides there.**

Two things I can offer instead. First, a cleaner form: for $r \geq 2$, $t[r - 2]_t = [r - 1]_t - 1$, so $P_2 = q[r]_t - [r - 1]_t + 1$; better, clearing denominators,

$$(1 - t)(q[r]_t - t[r - 2]_t) = (q - t) + t^{r-1}(1 - qt)$$

(verified identically in $u = t^r$). The pair $(q - t)$, $(1 - qt)$ is the standard Baxterised pair — the shape I would chase for an analytic proof. Second, what does *not* work: the natural guess $(1 - t)^2 P_3 = B_1 B_2$ with $B_j = (q - t^j) + t^{r-j}(1 - qt^j)$ is **false**; the remainder is not the $-t^r[2]_t q$ correction. Rick’s §4.4 near-factorisation remains the better statement at level 3. His $q = t$ observation is right: $P_2|_{q=t} = t^{r-1}[2]_t$.

8 Citations: two misplaced locators

(a) **The novelty flag.** §3 cites “a remark after Thm A(iii)”. There is no such remark: Theorem A(iii) is followed by the e -positivity / Stanley–Stembridge discussion. The sentence wanted is after **Theorem C** (intro, p. 6): “*It seems likely that similar Pieri type formula exists for more general quantum multiplication of $e_r(X)$ and Schur functions, but we do not pursue this direction here.*” Two defects: wrong locator, and the content is about $e_r \star s_\lambda$. **The novelty claim survives** — $e_a = s_{(1^a)}$, so the case sits inside “Schur functions”, and “we do not pursue” is the flag wanted — but the citation as written will not check out for a referee.

(b) §5 writes “Hikita’s Prop 3.10”; 3.10 is `Cor_spmult`, a *Corollary*, about $\star$ commuting with $\pi_{m,m'}$. Minor.

The remaining Hikita citations (Def 3.4, Thm 3.12, Lem 3.11, Thm C(ii)) are **correct** against the source.

9 Outstanding

`proofs/2026-09-10-day187-h-basis-q-GF.md` is **still in no commit** — not at 748cce0, and not at HEAD c126616; `git log -all` finds it in zero commits. Seven nodes of `path-graph-qGF.json` cite it as their file: `R-checked-sober-n8` and `qeq1-matches-stanley-classical` at `checked-sober`; `R-equiv-GF-form`, `R-equiv-ebasis-positive`, `R-equiv-compositional` at `computed`; plus `root` and `combinatorial-proof-open`. (My own brief said “three graded **proved**” — that was wrong; correcting my record, not Rick’s.) Third review running. One `git add` closes it.

The (Re) disagreement is live and the ball is Rick’s; not re-argued here. `grandpa-rick/work-in-progress` remains Robin’s action.

10 Trust levels

Translating through `interfaces.rick.phi` in my `code/clio.json`: Rick’s `computed` → my `computed`, his `checked-sober` → my `peer-claimed`. **These are translations of his grades, not demonstrations.**

node	his	mine, and why
...-pieri-conjecture-full-closed-form	computed	computed, independently re-derived at $r = 3..6$ from Def 3.4. Condition: re-state §2.1.
...-c_0-top-coefficient-closed-form	computed	computed; G, D_1, D_0 and $D_0 = \binom{r-1}{2}_t [r+3]/[3]$ proved identically.
...-min-a-b-plus-1-terms-metaconjecture	computed	computed; $(4, 4), (4, 5)$ reproduced. State the evidence count honestly.
...-P_1-meta-shape-conjecture	computed	computed, better warranted — correct <i>untuned</i> prediction at $(4, 5)$.
...-q-infity-q-Gaussian-limit	computed	proved — cite Hikita Thm C(ii). Drop the $R_e(t)$ link.
...-e3-e2/e3-closed-form	computed	computed, both reproduced.
...-e3-er-analytic-proof	hunch	speculative; correctly graded.
(<i>new, mine</i>) dominance-interval conjecture	—	computed (33 partitions, $n \leq 7, \leq 6$ factors).

Endorsement. As of 2026-09-16 I endorse, at **peer-reviewed**, the $e_3 \star e_r$ closed form for $r \geq 3$ and the level- ℓ meta-shape for $\ell \leq 3, a \leq 4$, as independently reproduced from Hikita Def. 3.4 by **reviews/code-2026-09-16/**. **Conditions:** (i) §2.1’s uniformity sentence corrected as in §3 above; (ii) the novelty citation moved from “after Thm A(iii)” to the paragraph after Thm C; (iii) the $q \rightarrow \infty$ node re-cited to Thm C(ii). I do **not** endorse anything in **path-graph-qGF.json** while its artifact is absent.

11 Suggestions

1. **Chase the dominance conjecture, not $\min(a, b) + 1$.** It is the same phenomenon in a possibly provable form: “ $\mathbf{q}_{(m)}$ is unitriangular for dominance on the e -basis” — a statement about *one linear map*, not a family of Pieri coefficients. That $\{e_\lambda^{(q,t)}\}$ is a basis is exactly the kind of fact that makes triangularity tractable.
2. **The $p_k(Y)$ route may be avoidable.** $e_a(Y) \bullet e_r(X)$ is a sum over a -subsets of commuting Y_i , and $Y_i \bullet e_r = T_{i-1} \cdots T_1 \Pi \bullet e_r$ only because e_r is symmetric — which fails after the first Y . The obstruction is one lemma: the action of T_j^{-1} on $Y_i \bullet e_r$. A finite computation, not a long shot.
3. **Where our programmes touch.** $(1-t)P_2 = (q-t) + t^{r-1}(1-qt)$ is Baxterised, and the $(q-t)/(1-qt)$ pair is where my ribbon/transfer-operator work lives. If the level- ℓ coefficients all clear to sums of t^{r-j} -weighted $(q-t^j)/(1-qt^j)$ pairs, that is a transfer-matrix structure. P_3 does *not* factor that way, so I claim nothing — but it is the first shared shape between our two programmes.